

Lahore Garrison University

DHA Phase 6 Lahore



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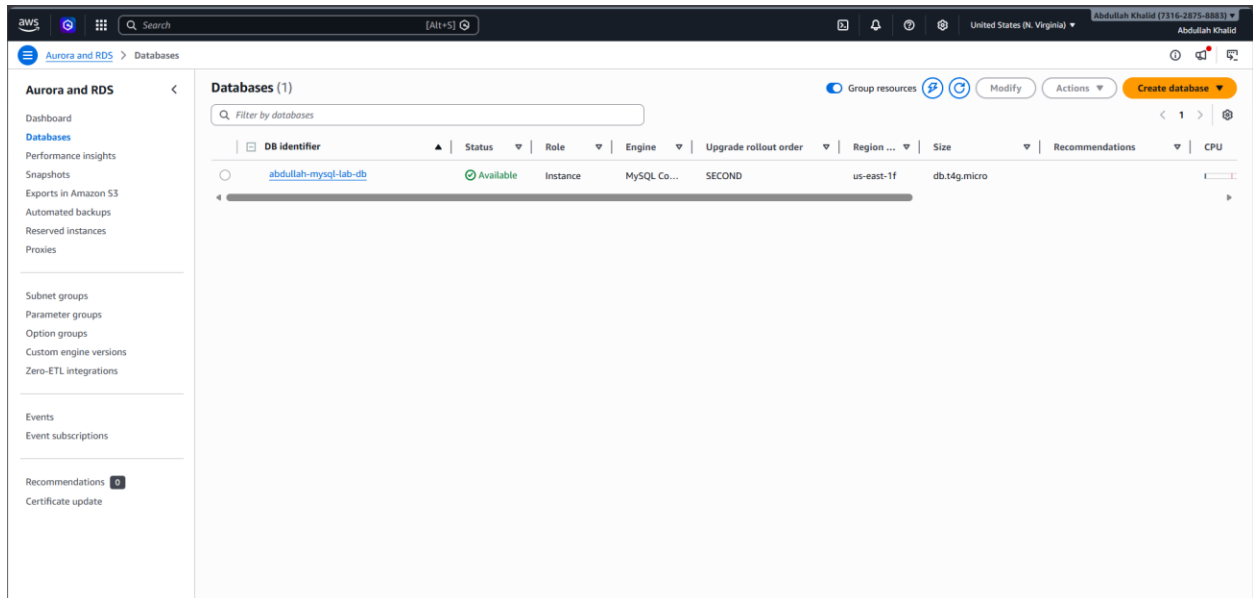
LAB 6

Database Management using Amazon RDS

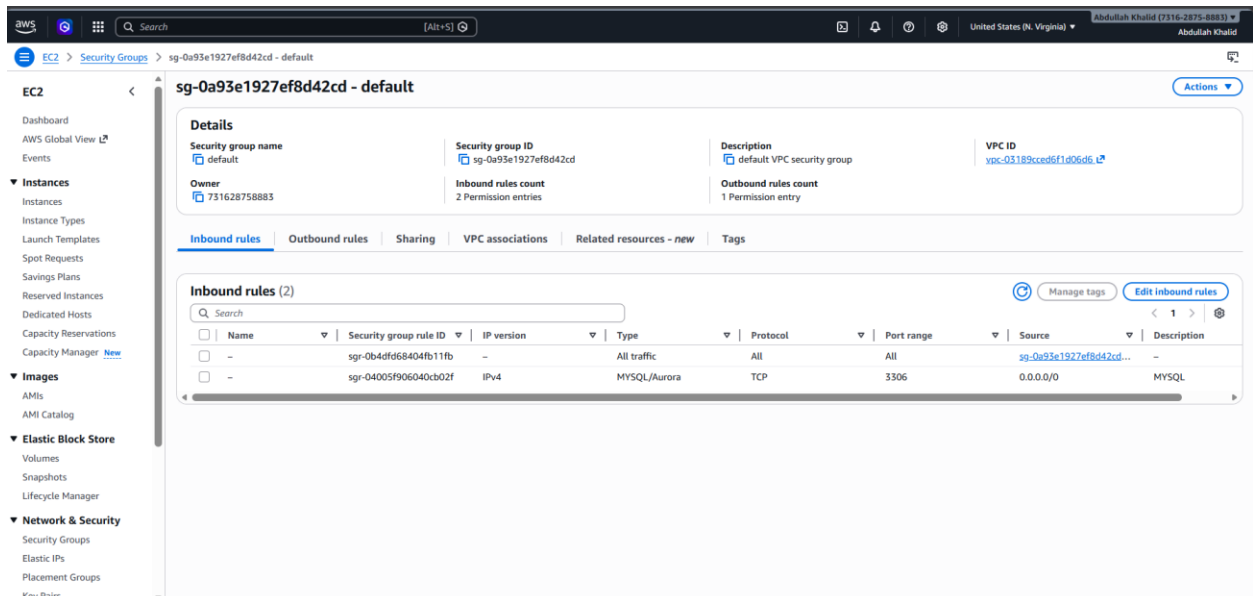
Question 1:

Create a database and insert multiple records.

Database Creation:



Security rule added to use Database in CloudShell:



Inserting Multiple Records in Database:

```
aws | [AWS Icon] | Search [Alt+S]

CloudShell

us-east-1 | +

~ $ mysql -h abdullah-mysql-lab-db.ck5waces0ayu.us-east-1.rds.amazonaws.com -P 3306 -u admin -pLabRDS123!
CREATE DATABASE labdb;
ERROR 2002 (HY000): Can't connect to MySQL server on 'abdullah-mysql-lab-db.ck5waces0ayu.us-east-1.rds.amazonaws.com' (115)
~ $ CREATE DATABASE labdb;
-bash: CREATE: command not found
~ $ mysql -h abdullah-mysql-lab-db.ck5waces0ayu.us-east-1.rds.amazonaws.com -P 3306 -u admin -pLabRDS123!

^[[A^[[[ERROR 2002 (HY000): Can't connect to MySQL server on 'abdullah-mysql-lab-db.ck5waces0ayu.us-east-1.rds.amazonaws.com' (115)
~ $
~ $ mysql -h abdullah-mysql-lab-db.ck5waces0ayu.us-east-1.rds.amazonaws.com -P 3306 -u admin -pLabRDS123!
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 42
Server version: 8.4.8 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> CREATE DATABASE labdb;
Query OK, 1 row affected (0.037 sec)

MySQL [(none)]> USE labdb;
Database changed
MySQL [labdb]> CREATE TABLE students(id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(50));
Query OK, 0 rows affected (0.049 sec)

MySQL [labdb]> INSERT INTO students(name) VALUES('Ali'),('Ahmed'),('Sara'),('Fatima');
Query OK, 4 rows affected (0.026 sec)
Records: 4  Duplicates: 0  Warnings: 0

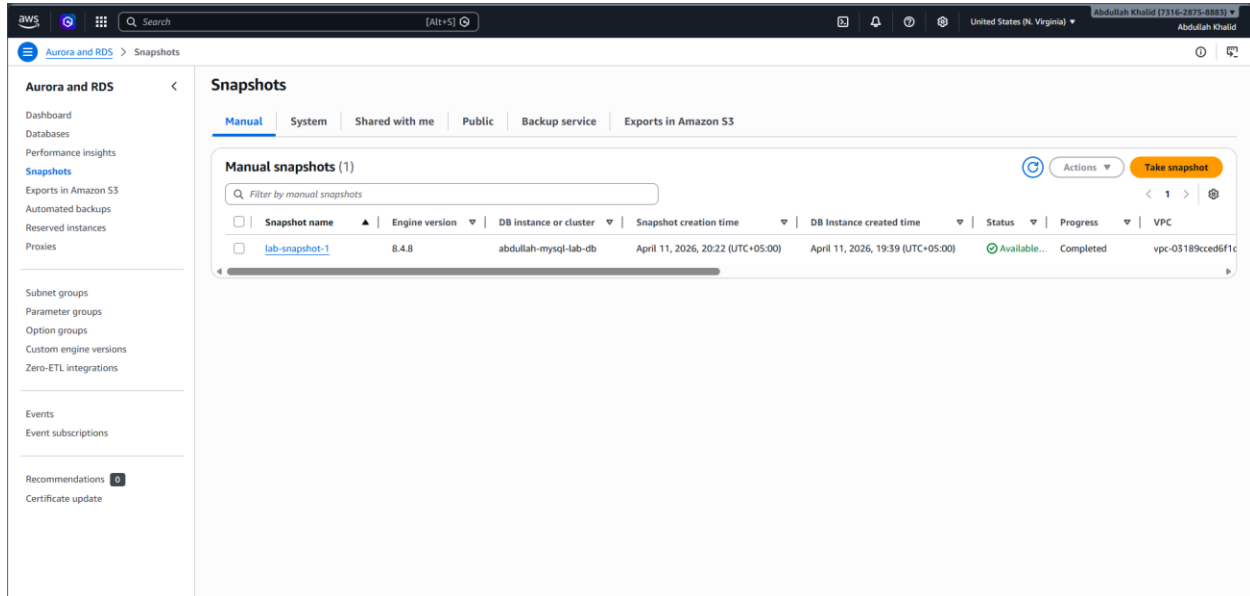
MySQL [labdb]> SELECT * FROM students;
+----+-----+
| id | name |
+----+-----+
| 1  | Ali  |
| 2  | Ahmed|
| 3  | Sara |
| 4  | Fatima|
+----+-----+
4 rows in set (0.002 sec)

MySQL [labdb]> █
```

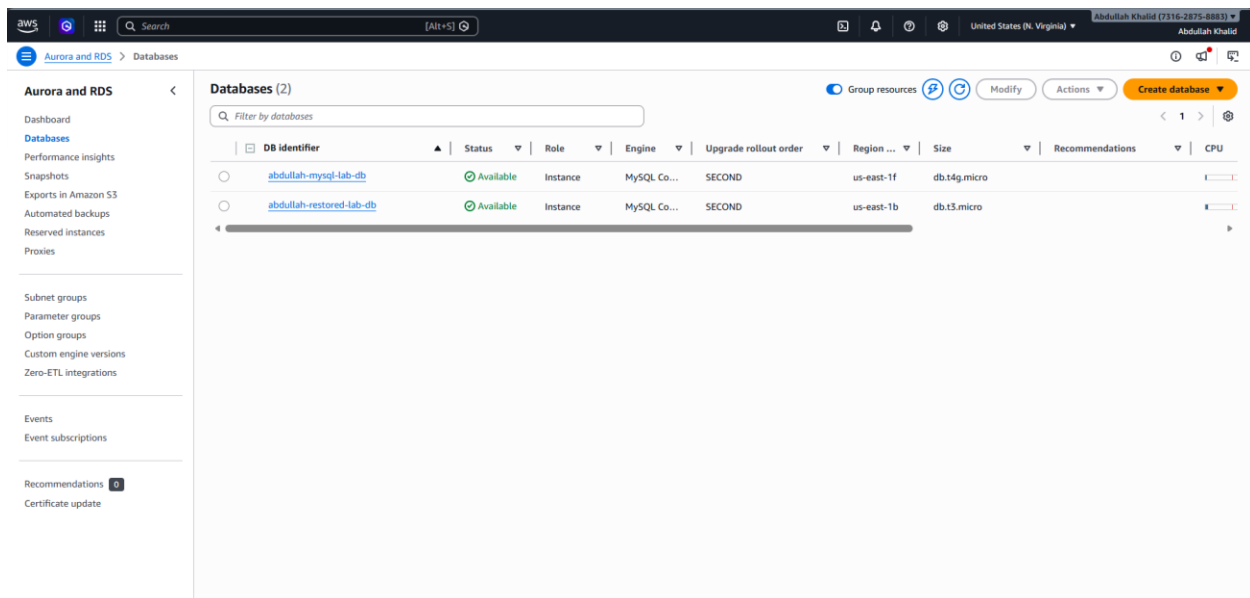
Question 2:

Take a snapshot and restore database:

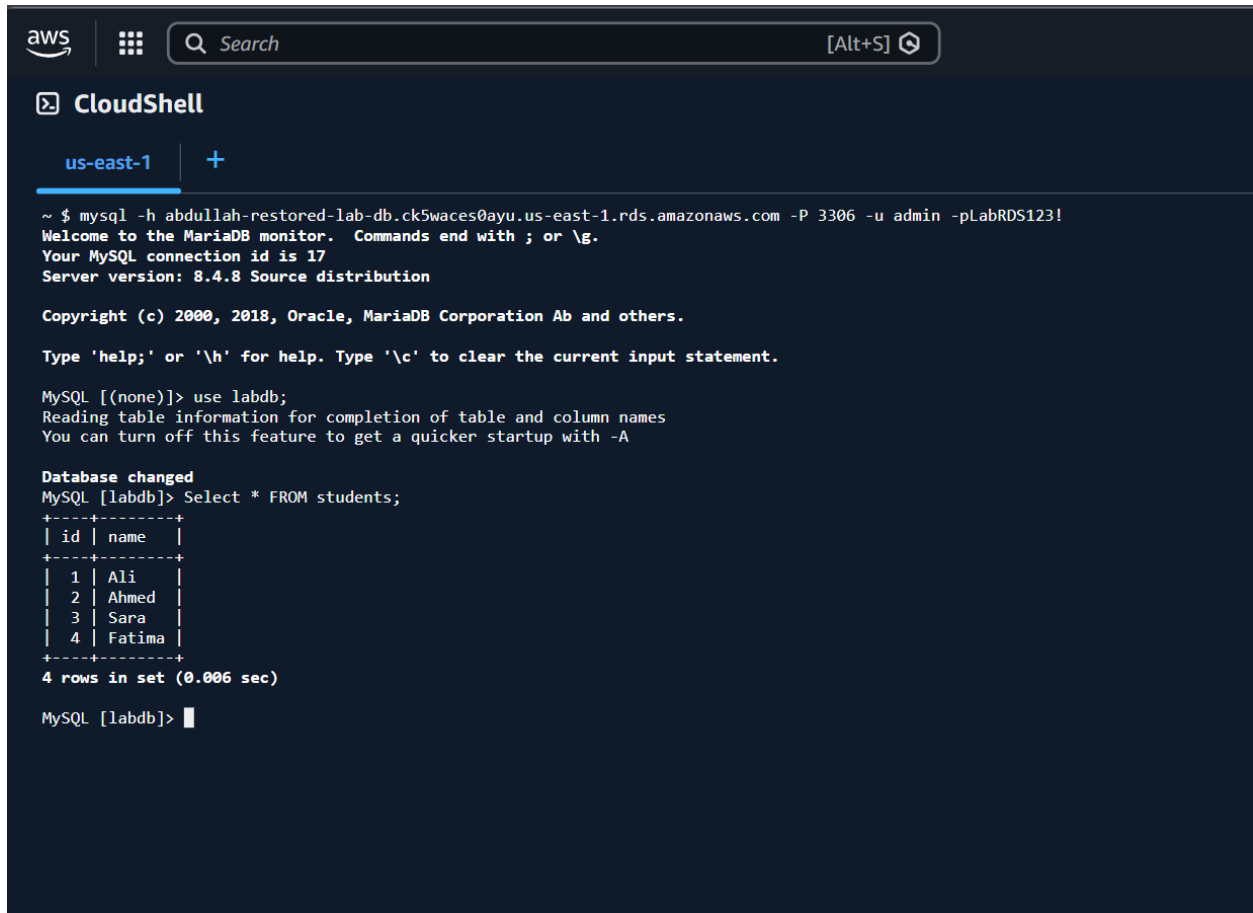
Taking Snapshot:



Restoring Database:



Same Data showing in this restored Database:



The screenshot shows the AWS CloudShell interface. At the top, there's an AWS logo, a grid icon, a search bar with the text "Search", and a keyboard shortcut "[Alt+S]". Below this, the "CloudShell" header is visible with a terminal icon and the text "us-east-1" followed by a plus sign. The terminal content shows a successful MySQL connection command, a welcome message, and the execution of a SQL query to select all records from a table named 'students'. The output is a table with 4 rows.

```
~ $ mysql -h abduallah-restored-lab-db.ck5waces0ayu.us-east-1.rds.amazonaws.com -P 3306 -u admin -pLabRDS123!
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 17
Server version: 8.4.8 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> use labdb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [labdb]> Select * FROM students;
+-----+
| id | name |
+-----+
| 1  | Ali  |
| 2  | Ahmed|
| 3  | Sara |
| 4  | Fatima|
+-----+
4 rows in set (0.006 sec)

MySQL [labdb]> 
```

Question 3:

Enable Multi-AZ and explain its benefit.

Benefits:

Production uses **db.t3.micro+** for:

- Automatic failover <60s if AZ fails
- Zero data loss (synchronous replication)
- No downtime during backups/patching

Multi-AZ unavailable on db.t4g.micro free tier:

The screenshot shows the AWS Management Console interface for configuring a MySQL DB instance. The instance type is set to **db.t4g.micro**. The storage type is **General Purpose SSD (gp2)** with 20 GiB allocated storage. The **Multi-AZ deployment** section shows the option **Do not create a standby instance** selected.

Instance type
db.t4g.micro
2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

Storage
Storage type: General Purpose SSD (gp2)
Allocated storage: 20 GiB
For high-throughput workloads, we recommend provisioning at least 100 GiB of General Purpose (SSD) storage. Lower values might result in higher latencies when the initial I/O credit balance is used up. [Learn more](#)

Availability & durability
Multi-AZ deployment: ☒ Do not create a standby instance